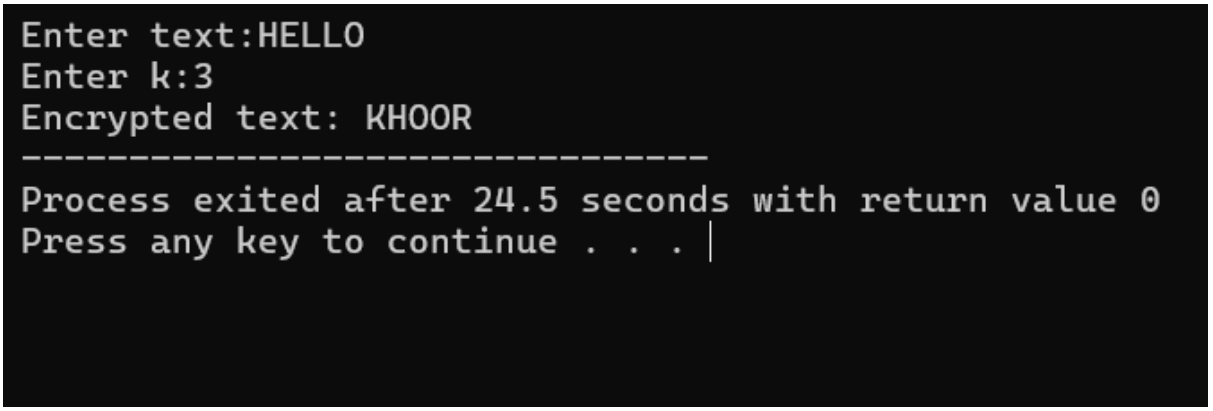
LAB EXPERIMENTS:

1.

```
#include<stdio.h>
int main()
{
	char s[100];
	int k,i;
	printf("Enter text:");
	scanf("%s",s);
	printf("Enter k:");
	scanf("%d",&k);
	for(i=0;s[i]!='\0';i++){
		s[i]=(s[i]-'A'+k)%26+'A';
	}
	printf("Encrypted text: %s",s);
	return 0;
}
```

OUTPUT:




2.

```
#include <stdio.h>
#include <ctype.h>

int main() {
   char text[200];
   char key[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
   int i, x;

   printf("Enter plaintext: ");
   fgets(text, sizeof(text), stdin);

   for (i = 0; text[i] != '\0'; i++) {
      if (isupper((unsigned char)text[i])) {
         x = text[i] - 'A';
```

```
            text[i] = key[x];
        } else if (islower((unsigned char)text[i])) {
            x = text[i] - 'a';
            text[i] = key[x] + ('a' - 'A');
        }
    }

    printf("Ciphertext: %s", text);
    return 0;
}
```

OUTPUT:

```
HELLO
OTDDH
--------------------------------
Process exited after 8.901 seconds with return value 0
Press any key to continue . . .
```

3.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char mat[5][5];

void build(char key[]) {
    int used[26] = {0}, i, j, r = 0, c = 0;
    char s[100];

    strcpy(s, key);
    strcat(s, "ABCDEFGHIKLMNOPQRSTUVWXYZ");

    for (i = 0; s[i]; i++) {
        char ch = toupper((unsigned char)s[i]);
        if (ch == 'J') ch = 'I';
        if (ch < 'A' || ch > 'Z' || used[ch-'A']) continue;
        used[ch-'A'] = 1;
        mat[r][c++] = ch;
        if (c == 5) { c = 0; r++; }
        if (r == 5) break;
    }
}

void encrypt(char a, char b) {
```

```
    int r1, c1, r2, c2, i, j;

    if (a == 'J') a = 'I';
    if (b == 'J') b = 'I';

    for (i=0; i<5; i++)
        for (j=0; j<5; j++) {
            if (mat[i][j] == a) { r1=i; c1=j; }
            if (mat[i][j] == b) { r2=i; c2=j; }
        }

    if (r1 == r2)
        printf("%c%c", mat[r1][(c1+1)%5], mat[r2][(c2+1)%5]);
    else if (c1 == c2)
        printf("%c%c", mat[(r1+1)%5][c1], mat[(r2+1)%5][c2]);
    else
        printf("%c%c", mat[r1][c2], mat[r2][c1]);
}

int main() {
    char key[100], text[200], s[200];
    int i, n = 0;

    printf("Enter keyword: ");
    scanf("%99s", key);
    printf("Enter plaintext: ");
    scanf(" %199[^\n]", text);

    build(key);

    for (i=0; text[i]; i++) {
        if (isalpha((unsigned char)text[i])) {
            char ch = toupper((unsigned char)text[i]);
            if (ch == 'J') ch = 'I';
            s[n++] = ch;
        }
    }
    s[n] = '\0';

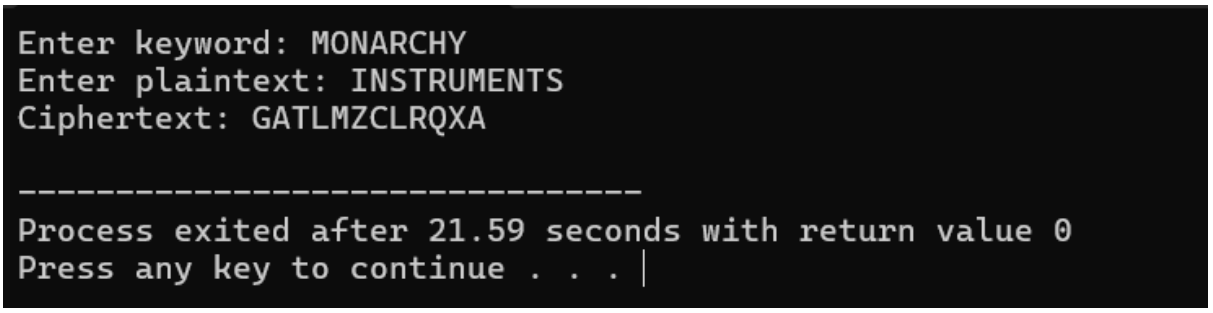
    printf("Ciphertext: ");
    for (i=0; i<n;) {
        char a = s[i++];
        char b;
        if (i == n) b = 'X';
        else if (s[i] == a) b = 'X';
        else b = s[i++];

        encrypt(a, b);
```

```
    }

    printf("\n");
    return 0;
}
```

OUTPUT:




4.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char text[200], key[100];
    int i, j = 0, k;

    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);
    printf("Enter key: ");
    scanf("%99s", key);

    for (i = 0; text[i] != '\0'; i++) {
        if (isalpha((unsigned char)text[i])) {
            char base = isupper((unsigned char)text[i]) ? 'A' : 'a';
            k = toupper((unsigned char)key[j % strlen(key)]) - 'A';
            text[i] = (text[i] - base + k) % 26 + base;
            j++;
        }
    }

    printf("Ciphertext: %s", text);
    return 0;
}
```

```
Enter plaintext: ATTACKATDAWN
Enter key: LEMON
Ciphertext: LXFOPVEFRNHR

--------------------------------
Process exited after 14.04 seconds with return value 0
Press any key to continue . . .
```

5.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int gcd(int a, int b) {
    while (b != 0) {
        int t = b;
        b = a % b;
        a = t;
    }
    return a;
}

int main() {
    char text[200];
    int a, b, i;

    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);
    printf("Enter a and b: ");
    scanf("%d%d", &a, &b);

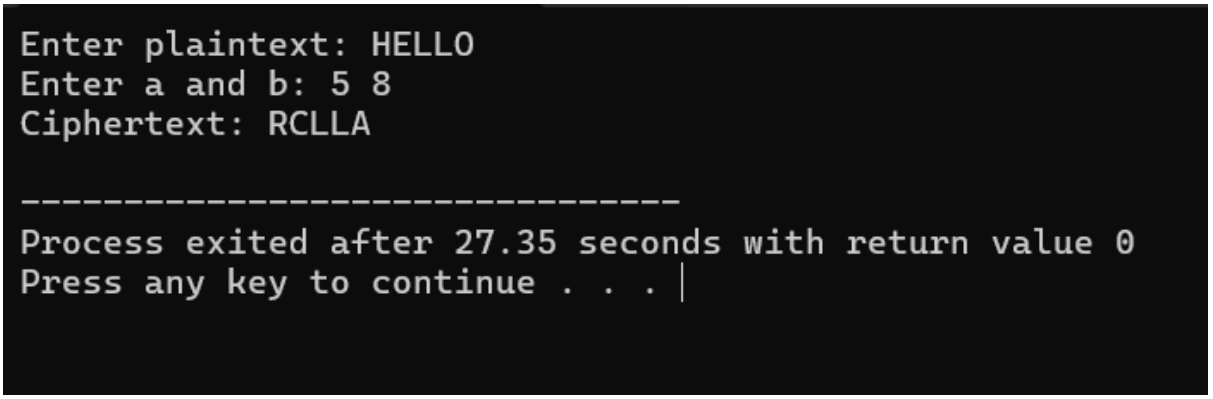
    a = (a % 26 + 26) % 26;
    b = (b % 26 + 26) % 26;

    if (gcd(a, 26) != 1) {
        printf("Invalid a value");
        return 0;
    }

    for (i = 0; text[i] != '\0'; i++) {
        if (isupper((unsigned char)text[i]))
            text[i] = ((a * (text[i]-'A') + b) % 26) + 'A';
        else if (islower((unsigned char)text[i]))
            text[i] = ((a * (text[i]-'a') + b) % 26) + 'a';
    }
```

```
    printf("Ciphertext: %s", text);
    return 0;
}
```

OUTPUT:




6.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int gcd(int a, int b) {
    while (b) {
        int t = b;
        b = a % b;
        a = t;
    }
    return a;
}

int main() {
    char c[500];
    int a, b, i, inv;

    printf("Enter ciphertext: ");
    fgets(c, sizeof(c), stdin);

    for (a = 1; a < 26; a++) {
        if (gcd(a, 26) != 1) continue;

        for (inv = 1; inv < 26; inv++)
            if ((a * inv) % 26 == 1) break;

        for (b = 0; b < 26; b++) {
            printf("a=%d b=%d: ", a, b);
```

```
            for (i = 0; c[i] != '\0'; i++) {
                if (isupper((unsigned char)c[i]))
                    putchar((inv * (c[i]-'A'-b+26)) % 26 + 'A');
                else if (islower((unsigned char)c[i]))
                    putchar((inv * (c[i]-'a'-b+26)) % 26 + 'a');
                else
                    putchar(c[i]);
            }
            printf("\n");
        }
    }
    return 0;
}
```

OUTPUT:

```
a=5 b=8: HELLO
```

7.

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char text[1000];
    int freq[26] = {0};
    int i, j, temp, index[26];

    printf("Enter ciphertext:\n");
    fgets(text, sizeof(text), stdin);

    for (i = 0; text[i]; i++)
        if (isalpha((unsigned char)text[i]))
            freq[toupper((unsigned char)text[i]) - 'A']++;

    for (i = 0; i < 26; i++)
        index[i] = i;

    for (i = 0; i < 25; i++)
        for (j = i + 1; j < 26; j++)
            if (freq[index[j]] > freq[index[i]]) {
                temp = index[i];
                index[i] = index[j];
                index[j] = temp;
            }
```

```
    printf("Letter frequencies:\n");
    for (i = 0; i < 26; i++)
        if (freq[index[i]] > 0)
            printf("%c = %d\n", 'A' + index[i],
                freq[index[i]]);

    return 0;
}
```

OUTPUT:

```
Enter ciphertext:
HELLO WORLD
Letter frequencies:
L = 3
O = 2
H = 1
D = 1
E = 1
R = 1
W = 1

--------------------------------
Process exited after 8.506 seconds with return value 0
Press any key to continue . . .
```

8.

```
#include <stdio.h>
#include <ctype.h>

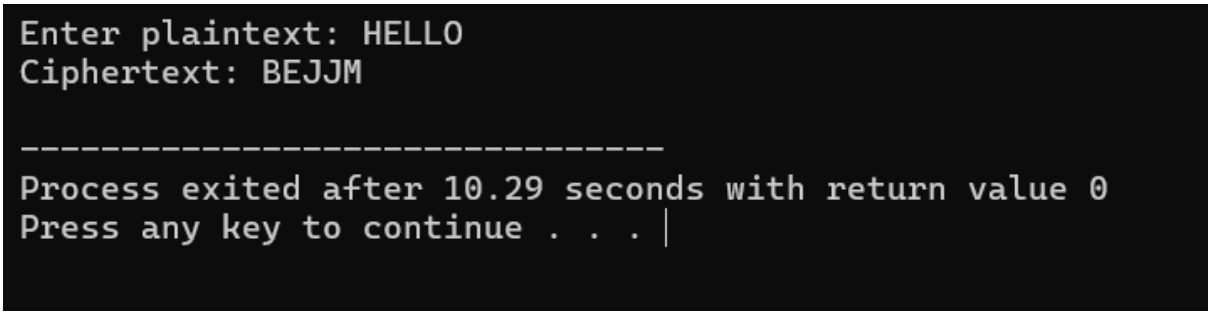
int main() {
    char text[200];
    char key[] = "CIPHERABDFGJKLMNOQSTUVWXYZ";
    int i, x;

    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);

    for (i = 0; text[i]; i++) {
        if (isupper((unsigned char)text[i])) {
            x = text[i] - 'A';
            text[i] = key[x];
        } else if (islower((unsigned char)text[i])) {
            x = text[i] - 'a';
            text[i] = key[x] + ('a' - 'A');
        }
    }
```

```
    printf("Ciphertext: %s", text);
    return 0;
}
```

OUTPUT:




9.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char mat[5][5];

void build(char key[]) {
    int used[26] = {0}, i, r = 0, c = 0;
    char s[200];

    strcpy(s, key);
    strcat(s, "ABCDEFGHIKLMNOPQRSTUVWXYZ");

    for (i = 0; s[i]; i++) {
        char ch = toupper((unsigned char)s[i]);
        if (ch == 'J') ch = 'I';
        if (ch < 'A' || ch > 'Z' || used[ch-'A']) continue;

        used[ch-'A'] = 1;
        mat[r][c++] = ch;
        if (c == 5) { c = 0; r++; }
        if (r == 5) break;
    }
}

void decrypt(char a, char b) {
    int r1=0,c1=0,r2=0,c2=0,i,j;

    if (a == 'J') a = 'I';
    if (b == 'J') b = 'I';
```

```
    for (i=0; i<5; i++)
        for (j=0; j<5; j++) {
            if (mat[i][j] == a) { r1=i; c1=j; }
            if (mat[i][j] == b) { r2=i; c2=j; }
        }

    if (r1 == r2)
        printf("%c%c", mat[r1][(c1+4)%5],
                    mat[r2][(c2+4)%5]);
    else if (c1 == c2)
        printf("%c%c", mat[(r1+4)%5][c1],
                    mat[(r2+4)%5][c2]);
    else
        printf("%c%c", mat[r1][c2], mat[r2][c1]);
}

int main() {
    char key[100], text[500];
    int i, n=0;
    char s[500];

    printf("Enter keyword: ");
    scanf("%99s", key);
    printf("Enter ciphertext: ");
    scanf("%499s", text);

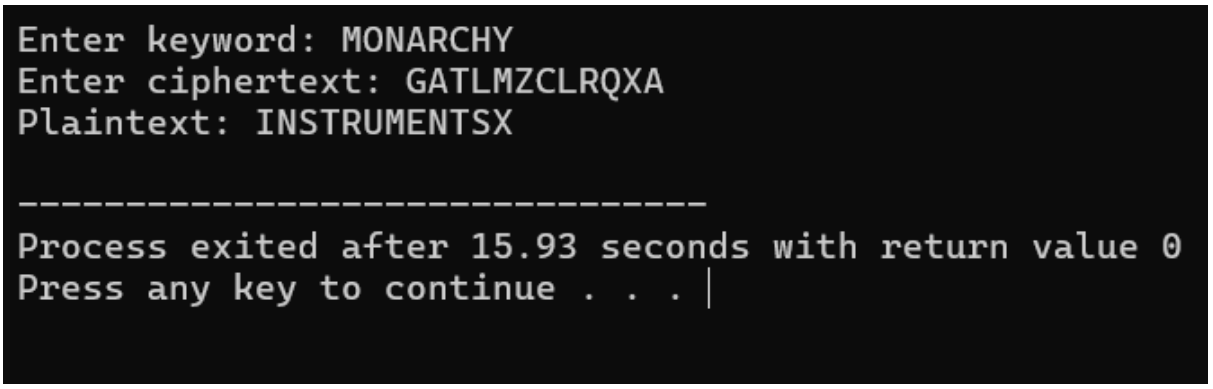
    build(key);

    for (i=0; text[i]; i++)
        if (isalpha((unsigned char)text[i]))
            s[n++] = toupper((unsigned char)text[i]);

    printf("Plaintext: ");
    for (i=0; i+1<n; i+=2)
        decrypt(s[i], s[i+1]);

    printf("\n");
    return 0;
}
```

OUTPUT:




10.

```
#include <stdio.h>
#include <ctype.h>

char mat[5][5] = {
   {'M','F','H','I','K'},
   {'U','N','O','P','Q'},
   {'Z','V','W','X','Y'},
   {'E','L','A','R','G'},
   {'D','S','T','B','C'}
};

void encrypt(char a, char b) {
   int r1=0,c1=0,r2=0,c2=0,i,j;

   if (a=='J') a='I';
   if (b=='J') b='I';

   for(i=0;i<5;i++)
      for(j=0;j<5;j++) {
         if(mat[i][j]==a) {r1=i;c1=j;}
         if(mat[i][j]==b) {r2=i;c2=j;}
      }

   if(r1==r2)
      printf("%c%c",mat[r1][(c1+1)%5],
                  mat[r2][(c2+1)%5]);
   else if(c1==c2)
      printf("%c%c",mat[(r1+1)%5][c1],
                  mat[(r2+1)%5][c2]);
   else
      printf("%c%c",mat[r1][c2],mat[r2][c1]);
}
```

```
int main() {
   char text[500], s[500];
   int i,n=0;

   printf("Enter plaintext: ");
   fgets(text,sizeof(text),stdin);

   for(i=0;text[i];i++)
      if(isalpha((unsigned char)text[i])) {
         char ch=toupper((unsigned char)text[i]);
         if(ch=='J') ch='I';
         s[n++]=ch;
      }

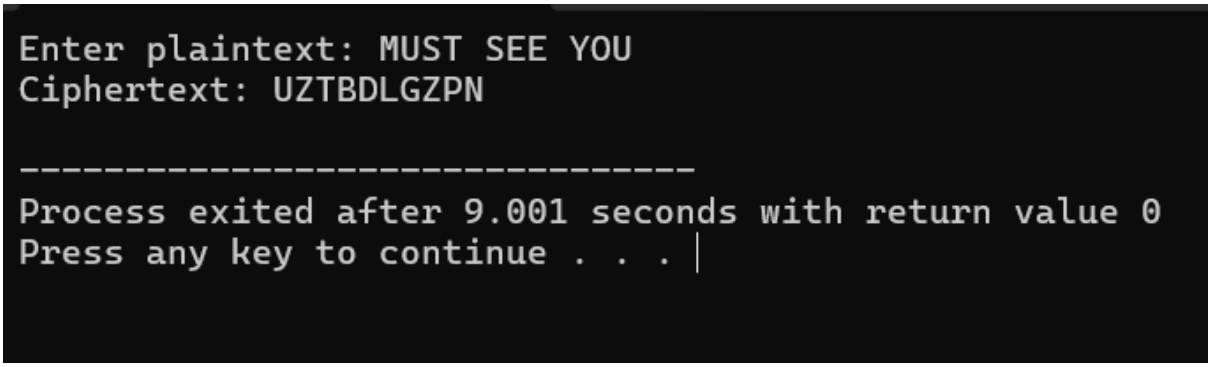
   printf("Ciphertext: ");
   for(i=0;i<n;) {
      char a=s[i++], b;

      if(i==n) b='X';
      else if(s[i]==a) b='X';
      else b=s[i++];

      encrypt(a,b);
   }

   printf("\n");
   return 0;
}
```

OUTPUT:




11.

```
#include <stdio.h>
#include <math.h>

int main() {
   long double keys = 1;
```

```
    int i;

    for (i = 1; i <= 25; i++)
        keys *= i;

    printf("Possible keys = 25! = %.0Lf\n", keys);
    printf("Approximate power of 2 = 2^%.2Lf\n",
            log2l(keys));

    return 0;
}
```

OUTPUT:

```
Possible keys = 25! = 0
Approximate power of 2 = 2^0.00

--------------------------------
Process exited after 0.07089 seconds with return value 0
Press any key to continue . . .
```

12.

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main() {
    char text[500], s[500];
    int i, n=0, a, b, c1, c2;

    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);

    for(i=0; text[i]; i++)
        if(isalpha((unsigned char)text[i]))
            s[n++] = toupper((unsigned char)text[i]);

    if(n % 2) s[n++] = 'X';

    printf("Ciphertext: ");

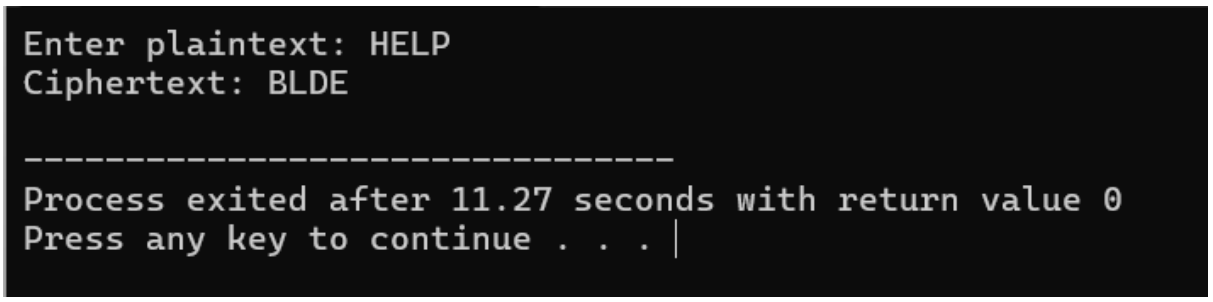
    for(i=0; i<n; i+=2) {
        a = s[i]-'A';
        b = s[i+1]-'A';
```

```
        c1 = (9*a + 4*b) % 26;
        c2 = (5*a + 7*b) % 26;

        printf("%c%c", c1+'A', c2+'A');
    }

    printf("\n");
    return 0;
}
```

OUTPUT:




13.

```
#include <stdio.h>

int main() {
    int a, b, c, d;

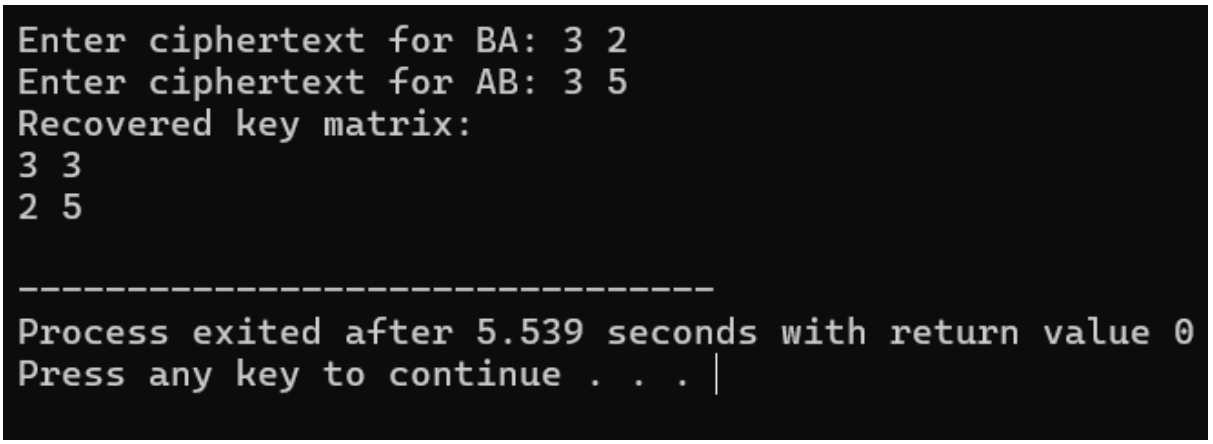
    printf("Enter ciphertext for BA: ");
    scanf("%d%d", &a, &c);

    printf("Enter ciphertext for AB: ");
    scanf("%d%d", &b, &d);

    printf("Recovered key matrix:\n");
    printf("%d %d\n", a, b);
    printf("%d %d\n", c, d);

    return 0;
}
```

OUTPUT:




14.

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main() {
   char text[] = "SENDMOREMONEY";
   char target[] = "CASHNOTNEEDED";
   int key[] = {9,0,1,7,23,15,21,14,11,11,2,8,9};
   int i, c;

   printf("Encrypted message: ");

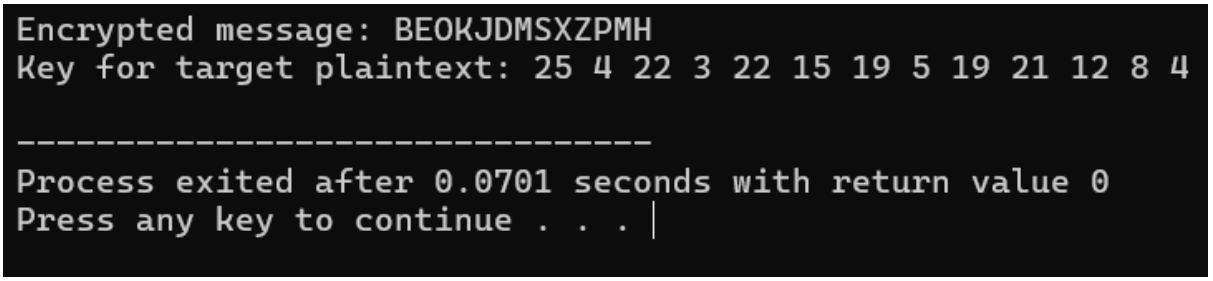
   for(i=0; i<13; i++) {
      c = (text[i]-'A'+key[i])%26;
      printf("%c", c+'A');
   }

   printf("\nKey for target plaintext: ");

   for(i=0; i<13; i++) {
      c = (text[i]-'A'+key[i])%26;
      printf("%d ", (c-(target[i]-'A')+26)%26);
   }

   printf("\n");
   return 0;
}
```

OUTPUT:




15.

```
#include <stdio.h>
#include <ctype.h>

int main() {
   char text[500];
   int k, i;

   printf("Enter ciphertext: ");
   fgets(text, sizeof(text), stdin);

   for(k=0; k<26; k++) {
      printf("Key %d: ", k);

      for(i=0; text[i]; i++) {
         if(isupper((unsigned char)text[i]))
            putchar((text[i]-'A'-k+26)%26+'A');
         else if(islower((unsigned char)text[i]))
            putchar((text[i]-'a'-k+26)%26+'a');
         else
            putchar(text[i]);
      }
      printf("\n");
   }

   return 0;
}
```

OUTPUT:

```
Enter ciphertext: KHOOR
Key 0: KHOOR

Key 1: JGNNQ

Key 2: IFMMP

Key 3: HELLO

Key 4: GDKKN

Key 5: FCJJM

Key 6: EBIIL

Key 7: DAHHK

Key 8: CZGGJ

Key 9: BYFFI

Key 10: AXEEH

Key 11: ZWDDG

Key 12: YVCCF

Key 13: XUBBE

Key 14: WTAAD

Key 15: VSZZC

Key 16: URYYB

Key 17: TQXXA

Key 18: SPWWZ

Key 19: ROVVY

Key 20: QNUUX

Key 21: PMTTW

Key 22: OLSSV

Key 23: NKRRU

Key 24: MJQQT

Key 25: LIPPS


--------------------------------
Process exited after 2.388 seconds with return value 0
Press any key to continue . . .
```